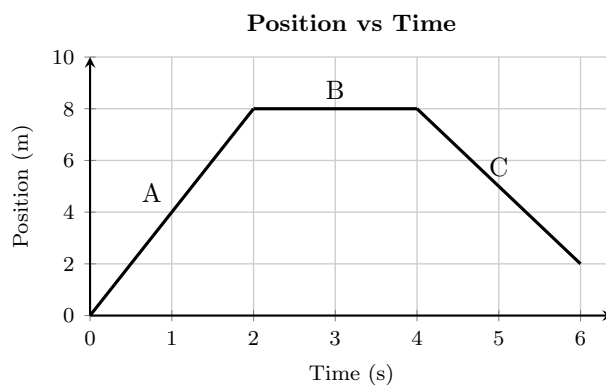


4. A car speeds up from 10 m/s to 30 m/s in 5 s. What is its acceleration?
5. A cyclist slows down from 12 m/s to 0 m/s in 4 s. What is the cyclist's acceleration?
6. A car drives around a circular track at a constant speed of 15 m/s. Is its velocity constant? Explain.
7. The position–time graph below shows the motion of a remote-controlled car along a straight track. The motion has three labeled segments, A, B, and C.



- (a) For each segment, state whether the velocity is positive, negative, or zero.

A: _____ B: _____ C: _____

- (b) Explain how the slope of each segment tells you the sign of the velocity.